
Closing the Algebra Is Not Enough: Lie-Closure Regularization Makes Learned Transformation Structure Comparable, Not Identifiable

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Abstract

When do independently trained models learn the same *transformation structure*, and when does agreement between them reflect the truth? We study this for transition-based representation learners that infer latent actions from pairs of observations and represent them with learned skew-symmetric generators. We add a Lie-closure penalty (BracketNet) and evaluate on $\text{SO}(2)$, T^2 and $\text{SO}(3)$ actions observed through nonlinear maps and through rendered images, with every choice fixed on development seeds and fresh seeds evaluated once.

(1) Transition fitting alone recovers the abelian T^2 algebra in most runs, but leaves the $\text{SO}(3)$ generator span non-closed in 15/20 runs. Closure regularization closes it, roughly halves the distance to the true algebra, and makes the algebras of independently trained models agree more (Holm-corrected $p \leq 8.8 \times 10^{-3}$), with no detectable change in transport error. (2) Closed is not correct. 8/20 $\text{SO}(3)$ runs close onto a *chiral* $\mathfrak{su}(2)$, and on rendered images every closed run (7/20) is chiral. We prove a partial explanation: a rotation about the axis through a point leaves that point fixed, so the true action cannot be inferred from endpoints and no continuous endpoint rule makes the true algebra exact, whereas a chiral algebra acts simply transitively and admits an exact rule. Composition errors, a pre-registered intervention with true action coefficients, and a generator-span trace agree with this account. (3) Transfer of transformations between models tracks whether they learned the *same* algebra, even a wrong one. Linear CKA catches encoder failures but not these class mismatches.

Algebraic closure turns continuous disagreement between models into disagreement among a few closed subalgebras: it makes learned transformation structure comparable, but not identifiable.

1 Introduction

Representation-similarity methods ask whether two networks embed data in the same way, up to a class of transformations [Kornblith et al., 2019, Williams et al., 2021, Moschella et al., 2023]. For models that learn how data *transform*, there is a second question: do they agree on the transformations themselves? This matters when actions inferred by one model are to be used in another, and when a learned symmetry is read as a scientific statement. Embedding agreement does not settle it. Two models can embed the same data almost identically and still represent its transformations by different Lie algebras, and two models can agree perfectly with each other while both being wrong (Cor. 1).

We study this in transition-based symmetry learning [Park et al., 2022, Keurti et al., 2023, Yang et al., 2024]. A model infers a latent action from a pair of observations and represents it through learned generators. Nothing in a transition loss requires those generators to span a Lie algebra, so a natural structural bias is to penalize commutators that leave their span. We ask: *when do independently learned transformation representations become structurally comparable, and why does algebraic consistency fail to make them identifiable?*

Contributions.

- **An evaluation that separates five levels of agreement:** embedding, algebra, representation class, inferred action and ground truth. It includes cross-model transformation transfer, uses disjoint seed pairs as independent units, and pre-registers its primary tests.
- **Evidence that closure changes the *kind* of disagreement.** For $SO(3)$, transition fitting leaves generator spans non-closed and scattered. A closure penalty concentrates them on the three closed classes allowed in $\mathfrak{so}(4)$, one correct and two chiral, which improves cross-model agreement without guaranteeing correctness.
- **A mechanism with a proof** (Prop. 3). Endpoint-based action inference cannot capture rotations about the axis through a point, and no continuous endpoint rule makes the true algebra exact, whereas a chiral algebra admits an exact rule. The result is qualitative (no quantitative gap). Three tests, two of them pre-registered, support this account.
- **A transfer analysis:** actions transfer between models that learned the same algebra, right or wrong. CKA catches representation failures but not algebra-class mismatches.

We also give a corrected invariance analysis of the closure loss and a local bound on how far products of group elements leave the learned span. The underlying Lie theory is standard; we cite it as such.

2 Related work

Learning transformation groups. Lie-group transformation models have a long history [Rao and Ruderman, 1999, Sohl-Dickstein et al., 2010, Cohen and Welling, 2014], including transport operators in autoencoder latents [Connor and Rozell, 2020]. Recent methods discover symmetries from predictors [Dehmamy et al., 2021], adversarially from distributional invariance [Yang et al., 2023], in learned latent spaces [Yang et al., 2024], beyond affine families [Shaw et al., 2024], or as one-parameter families [Gabel et al., 2023]. Closure losses that force learned generators to span a (sub)algebra already exist [Forestano et al., 2023], including studies of $\mathfrak{so}(4)$ subalgebras; our closure term is an instance of this idea. Equivariant networks assume a known group [Cohen and Welling, 2016, Bronstein et al., 2021].

Group-structured representations from transitions. Symmetric embedding networks [Park et al., 2022] use known actions. Homomorphism autoencoders [Keurti et al., 2023] learn a homomorphism from action labels into a matrix group without knowing the group. Symmetry-based disentanglement [Higgins et al., 2018] defines representations through group actions and needs interaction [Caselles-Dupré et al., 2019]. Rotation representations of an agent’s actions have been learned from interaction sequences [Quessard et al., 2020], and actions have been estimated without labels [Painter et al., 2020], and group decompositions have been shown identifiable under assumptions [Dang-Nhu et al., 2026]. We share the transition setting and, like Painter et al. [2020], use *no* action labels. Our oracle-coefficient intervention (Sec. 6.2) is the action-supervised counterpart and shows what the missing labels cost. LieGAN and LaLiGAN learn from distributional signals rather than transitions, so we do not compare against them directly.

Comparing representations. CKA [Kornblith et al., 2019], shape metrics [Williams et al., 2021] and relative representations [Moschella et al., 2023] compare embeddings up to a transformation class, and convergence of representations across models is an active hypothesis [Huh et al., 2024]. These tools compare *point clouds*. We compare the transformation algebras acting on them, and the transfer of actions between models.

What is not new. Closure penalties, the classification of subalgebras of $\mathfrak{so}(4)$ and $\mathfrak{so}(5)$, Dynkin’s formula, the hairy-ball theorem and quaternion transitivity are standard. Topological obstructions to

continuous encoders and rotation representations are known [de Haan and Falorsi, 2018, Zhou et al., 2019, Bouchacourt et al., 2021]; Prop. 3b is an obstruction of the same kind, applied to endpoint action inference. Our contribution is their consequences for this learning problem and the empirical analysis. Appendix D compares settings in detail.

3 Setting and method

Observations $x = h(u)$ come from states u through an unknown injective map. Training data are unlabeled triples $(x_0, x_1, x_2) = (h(u), h(g_1 u), h(g_2 g_1 u))$ with near-identity g_i from an unknown compact group, and K is given.

The model has an encoder E , a decoder D , an action-inference network $a = q(z, z')$ (bounded by $0.65 \tanh$) and generators $A_k \in \mathfrak{so}(d)$, each rescaled to $\|A_k\|_F = \sqrt{2}$. An action is $\rho(a) = \exp(\sum_k a_k A_k)$.

With $z_t = E(x_t)$ and $a_{st} = q(z_s, z_t)$, the losses are:

- reconstruction $\sum_t \|D(z_t) - x_t\|^2$;
- transport in latent and observation space, $\sum_{t < 2} \|\rho(a_{t,t+1})z_t - z_{t+1}\|^2$ and $\sum_{t < 2} \|D(\rho(a_{t,t+1})z_t) - x_{t+1}\|^2$;
- composition $\|\rho(a_{02}) - \rho(a_{12})\rho(a_{01})\|_F^2$;
- a Gram penalty $\|B^\top B - 2I\|_F^2$ and an anti-collapse term $\|\text{Cov}(z) - I\|_F^2$;
- the closure residual

$$\mathcal{L}_{\text{close}} = \frac{1}{K^2} \sum_{i,j} \|(I - P_\varepsilon) \text{vec}[A_i, A_j]\|_2^2, \quad P_\varepsilon = B(B^\top B + \varepsilon I)^{-1} B^\top, \quad (1)$$

where B stacks the vectorized generators.

LOCAL uses reconstruction, transport and the regularizers. +COMP. adds composition (weight 1.2). BRACKETNET adds $\lambda(t)\mathcal{L}_{\text{close}}$: $\lambda(t) = 0$ for 35% of updates, a linear ramp to λ by 70%, and frozen generators afterwards. All three share networks, data and optimizer (weights in Appendix B).

4 Theory: what closure guarantees, and why it can prefer the wrong algebra

Notation: $\mathfrak{a} = \text{span}(A)$, $G = B^\top B$, P_0 the orthogonal projector onto $\mathfrak{a}$, $N = I - P_0$. Proofs are in Appendix A. Standard Lie-theoretic facts are cited as such; our contributions are their consequences for this learning objective.

Invariances. With $\varepsilon=0$ and $M \in GL(K)$, $\mathcal{L}_{\text{close}}(BM) = K^{-2} \|R(M \otimes M)\|_F^2$, where R is the residual matrix, so $\sigma_{\min}(M)^4 \mathcal{L}_{\text{close}} \leq \mathcal{L}_{\text{close}}(BM) \leq \sigma_{\max}(M)^4 \mathcal{L}_{\text{close}}$. The loss is invariant only under orthogonal reparameterizations; its zero set (“ $\mathfrak{a}$ is a subalgebra”) is $GL(K)$ -invariant. For $\varepsilon > 0$, $P_\varepsilon(BM) = B(G + \varepsilon M^{-\top} M^{-1})^{-1} B^\top$. Because $\text{diag } G = 2$, the Gram target affects no gradient.

Proposition 1 (Closure, escape and integration). (a) If $A_1, \dots, A_K$ are independent and $[A_i, A_j] \in \mathfrak{a}$, then $\mathfrak{a}$ is a Lie subalgebra and integrates to a connected immersed subgroup of $SO(d)$ (standard). (b) The uniform escape constant $\delta = \sup_{X,Y \in \mathfrak{a}} \|N[X, Y]\|_F / (\|X\|_F \|Y\|_F)$ satisfies $\delta \leq K \sqrt{\mathcal{L}_{\text{close}}(A; 0) / \lambda_{\min}(G)}$. (c) For $X, Y \in \mathfrak{a}$ with $\|X\|_F, \|Y\|_F \leq r < \frac{\ln 2}{4}$, $\text{dist}(\log(e^X e^Y), \mathfrak{a}) \leq \frac{\delta}{4} [-\ln(2 - e^{4r}) - 4r] = \frac{1}{2} \delta r^2 + O(\delta r^3)$. (d) Chained orthogonal transport accumulates per-step errors at most additively.

Part (c) is local: it covers rotations of up to ≈ 0.12 rad, smaller than typical inferred actions.

Closed subalgebras in the benchmarks. Write $\mathfrak{so}(4) = \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ (two commuting ideals).

Proposition 2 (Classification; standard facts applied to our benchmarks). (a) Every 2-dimensional subalgebra of $\mathfrak{so}(5)$ is abelian and $SO(5)$ -conjugate to the true $\mathfrak{t}^2$. (b) Every 3-dimensional subalgebra of $\mathfrak{so}(4)$ is isomorphic to $\mathfrak{su}(2)$. It is $SO(4)$ -conjugate to exactly one of three: the diagonal

125 $\mathfrak{so}(3)$ acting on $\mathbb{R}^3 \oplus \mathbb{R}$ (the truth), $\mathfrak{su}(2)_L$ or $\mathfrak{su}(2)_R$. Under $O(4)$ the two chiral factors merge into
 126 one class. (c) No closed 4-dimensional span in $\mathfrak{so}(4)$ contains the diagonal $\mathfrak{so}(3)$.

127 The three $SO(3)$ classes are separated by the Pfaffian of unit elements $(0, +\frac{1}{2}, -\frac{1}{2})$ and by the
 128 participation ratio (2 vs. 4). Diagonal and chiral spans are at subspace distance exactly $\frac{1}{2}$ (principal
 129 angles of 45°). An optimization that does not use the classification confirms it: minimizing a span-
 130 only closure residual from 200 random 3-dimensional subspaces of $\mathfrak{so}(4)$ converged to 192 diagonal
 131 and 8 chiral spans, and to nothing else.

132 **Endpoint ambiguity: why the objective can favour a wrong closed algebra.** BracketNet infers
 133 each action from the two endpoint codes, $a = q(z, z')$. For the true $SO(3)$ action this is ill-posed.
 134 A rotation about the axis through z fixes z , so g and $g \exp(t\hat{z})$ have the same endpoints. A chiral
 135 $\mathfrak{su}(2)$ has no such ambiguity: left multiplication by unit quaternions acts simply transitively on every
 136 sphere of $\mathbb{R}^4 \cong \mathbb{H}$. Call an endpoint rule q *exact* on a triple (z_0, z_1, z_2) if $\rho(q(z_i, z_j))z_i = z_j$ for
 137 $ij \in \{01, 12, 02\}$ and $\rho(q(z_0, z_2)) = \rho(q(z_1, z_2))\rho(q(z_0, z_1))$.

138 **Proposition 3** (Endpoint ambiguity). *On the $SO(3)$ benchmark, use the ground-truth state as the*
 139 *latent code, with an exact decoder. (a) With chiral generators, $q(z, z') = \log(z'z^{-1})$ is exact on*
 140 *every triple on a sphere $\|z\| = \text{const} > 0$ (all data triples); it is smooth wherever $z' \neq -z$. The*
 141 *closure, Gram and covariance terms vanish too. Exactness fails only where a coefficient exceeds*
 142 *the bound. (b) With the true generators, for every $\epsilon > 0$ and every orbit sphere $\{(v, z_4) : \|v\| = c\}$,*
 143 *$c > 0$, no continuous rule is exact on all triples in that sphere with pairwise angles $< \epsilon$. (c) For*
 144 *any action-inference rule based on endpoints, $\mathbb{E}\|\rho(q(z, gz)) - g\|^2 > 0$ under the true action: the*
 145 *stabilizer component of g is not identifiable from (z, gz) .*

146 Part (b) follows from the hairy-ball theorem: exactness on small triangles would define a continuous
 147 section of $SO(3) \rightarrow S^2$. It concerns *exact* zero for *continuous* rules only. A discontinuous rule can
 148 be exact off a null set, and we do not bound how close continuous rules can come (App. A, Remark).
 149 On the data, ignoring the coefficient bound, the minimal-rotation rule for the true algebra transports
 150 exactly but leaves a composition residual of 3.2×10^{-3} ; the chiral rule leaves $\sim 10^{-31}$.

151 **Corollary 1** (Agreement without correctness). *Two models with identical ground-truth embeddings*
 152 *and $\mathfrak{su}(2)_L$ generators have linear CKA 1 with each other and with the true state, zero closure,*
 153 *transport and composition loss (up to coefficient clipping), and cross-model algebra distance 0, but*
 154 *distance $\frac{1}{2}$ to the true algebra and inferred actions that are not the true rotations.*

155 **The implication chain, made precise.** Write (i) closure, (ii) correct conjugacy class, (iii) recovery
 156 after latent alignment, (iv) correct action inference. (i) $\Rightarrow$ (ii) holds for T^2 (Prop. 2a) and fails for
 157 $SO(3)$ (Prop. 2b). (ii) $\Rightarrow$ (iii) also needs an encoder close to an isometry. (iv) is unattainable for
 158 $SO(3)$ by every endpoint method, even with the true algebra (Prop. 3c). Agreement between models
 159 implies none of (ii)–(iv) (Cor. 1). Proposition 3 describes the loss landscape at one embedding, and
 160 qualitatively. It does not show that the minimum over trainable rules is lower for chiral solutions, nor
 161 which solution gradient descent reaches; Sec. 6.2 tests the latter empirically.

162 5 Experimental protocol

163 **Benchmarks.** $SO(2)$, T^2 and $SO(3)$ act linearly on a latent state in $\mathbb{R}^d$ that includes an invariant
 164 coordinate ($d = 3, 5, 4$). The true algebra is therefore a proper subspace of $\mathfrak{so}(d)$, and random spans
 165 do not close: the mean closure of random normalized spans is 0.60 ($K=2$ in $\mathfrak{so}(5)$) and 0.66 ($K=3$
 166 in $\mathfrak{so}(4)$).

167 States are drawn as $z_0 \sim \mathcal{N}(0, I)$, and transitions apply $\exp(\sum_k a_k T_k)$ with $a_k \sim \mathcal{N}(0, 0.24^2)$.
 168 Observations are $x = Qz + 0.18 \tanh(Bz)$, one fixed injective map per group. The rendered
 169 benchmark is described in Sec. 6.3. Each run trains on 1,400 triples and is evaluated on 500 ten-edge
 170 sequences.

171 **Selection on development seeds only.** We fixed the training budget, closure weight and closure
 172 objective on development seeds 5–9, with rules committed before those runs:

- 173 • *Budget:* the smallest budget at which +COMP. transport is within 10% of its value at 4,000
 174 updates. This rule is method-agnostic.

Table 1: Fresh-seed results (seeds 100–119, mean $\pm$ s.d.; frozen protocol). GT dist.: ground-truth algebra distance after Procrustes alignment (chance 0.80 for T^2 , 0.50 for SO(3)). CC: correct-closed runs. SO(2), where all methods coincide, is in Appendix Table 4.

Group	Method	1-step MSE	10-step MSE	Closure residual	GT dist.	CC
T^2	Local	0.0080 $\pm$ 0.0076	0.0645 $\pm$ 0.0841	$3.4 \times 10^{-1} \pm 7.2 \times 10^{-1}$	0.161 $\pm$ 0.282	14/20
T^2	+Comp.	0.0081 $\pm$ 0.0076	0.0605 $\pm$ 0.0757	$2.9 \times 10^{-1} \pm 6.3 \times 10^{-1}$	0.151 $\pm$ 0.274	14/20
T^2	BracketNet	0.0072 $\pm$ 0.0079	0.0539 $\pm$ 0.0792	$2.2 \times 10^{-2} \pm 5.6 \times 10^{-2}$	0.129 $\pm$ 0.282	16/20
T^2	BracketNet ($\lambda=0.1$)	0.0082 $\pm$ 0.0078	0.0622 $\pm$ 0.0785	$3.1 \times 10^{-1} \pm 6.1 \times 10^{-1}$	0.157 $\pm$ 0.277	14/20
SO(3)	Local	0.0048 $\pm$ 0.0027	0.0325 $\pm$ 0.0312	$6.3 \times 10^{-1} \pm 5.7 \times 10^{-1}$	0.479 $\pm$ 0.435	6/20
SO(3)	+Comp.	0.0048 $\pm$ 0.0031	0.0328 $\pm$ 0.0299	$6.2 \times 10^{-1} \pm 5.7 \times 10^{-1}$	0.475 $\pm$ 0.433	5/20
SO(3)	BracketNet	0.0049 $\pm$ 0.0034	0.0319 $\pm$ 0.0341	$2.5 \times 10^{-2} \pm 7.1 \times 10^{-2}$	0.243 $\pm$ 0.254	8/20
SO(3)	BracketNet ($\lambda=0.1$)	0.0050 $\pm$ 0.0031	0.0484 $\pm$ 0.0484	$5.4 \times 10^{-1} \pm 5.4 \times 10^{-1}$	0.453 $\pm$ 0.413	5/20

Table 2: Pre-registered primary hypotheses, BracketNet vs. +COMP. (one-sided). Units: seeds ($n=20$) for H1–H2, disjoint seed pairs ($n=10$) for H3. Reported: two-sided 95% t -interval of the difference, paired effect size d_z , and the number of units where BracketNet is lower. p : exact sign-flip; Holm over six tests.

Hypothesis	Group	n	+Comp.	BracketNet	95% CI (t)	d_z	lower	p	Holm p
H1 closure	T^2	20	0.2906	0.0216	[-0.5427, +0.0049]	-0.46	16/20	4.3e-05	2.1e-04
H2 ground-truth distance	T^2	20	0.1506	0.1289	[-0.0674, +0.0240]	-0.22	16/20	1.6e-01	3.1e-01
H3 cross-seed distance	T^2	10	0.2138	0.1981	[-0.1421, +0.1106]	-0.09	7/10	3.8e-01	3.8e-01
H1 closure	SO(3)	20	0.6163	0.0250	[-0.8550, -0.3278]	-1.05	20/20	9.5e-07	5.7e-06
H2 ground-truth distance	SO(3)	20	0.4754	0.2429	[-0.3348, -0.1301]	-1.06	19/20	1.2e-04	4.9e-04
H3 cross-seed distance	SO(3)	10	0.6845	0.3922	[-0.4424, -0.1423]	-1.39	9/10	2.9e-03	8.8e-03

• *Closure weight*: the λ with the lowest closure among those whose transport is not worse than +COMP.’s. This rule needed an amendment, committed before any fresh run: no λ met the constraint, so the tolerance was relaxed in steps of 10%, and $\lambda = 3$ qualified at 1.20.

The result is 4000 updates, $\lambda = 3$ and Eq. (1). All runs evaluate the final iterate. $\lambda = 0.1$, the originally planned weight, is evaluated at the same budget as a reference.

Seeds and pre-registration. Main evaluation: 20 fresh seeds (100–119). Ablations: 200–209. Rendered benchmark: 300–319. Oracle intervention: 400–419. Each set was used once. Structural thresholds, primary hypotheses and the predictions for the mechanism and transfer experiments were committed before the corresponding runs.

Structural categories (fixed in advance). A run is *collapsed* if $\sigma_{\min}/\sigma_{\max}(B) < 0.1$, and *closed* if its closure is at most 1% of the random-span level. It is *correct-closed* if it is closed and its ground-truth algebra distance (mean squared sine of the principal angles between $\text{span}(RAR^T)$ and the true span, with R the Procrustes map from codes to true states) is at most 10% of chance. A closed SO(3) span with participation ratio ≥ 3 is *chiral*.

Statistics. Seeds are the independent units; cross-seed quantities use the 10 *disjoint* seed pairs. The primary family compares BracketNet with +COMP., one-sided, with exact paired sign-flip tests and Holm correction: closure (H1), ground-truth algebra distance (H2) and cross-seed algebra distance (H3), each for T^2 and SO(3). Everything else is secondary, and we call a difference detectable only if its 95% interval excludes zero.

6 Results

6.1 Closure makes SO(3) algebras closed, but often wrong

Transition fitting closes T^2 but not SO(3). Without closure regularization, the true T^2 algebra is recovered in 14/20 (+Comp.) and 14/20 (Local) runs, while the SO(3) span stays *non-closed* in 15/20 and 14/20 runs. One- and ten-step transport errors are low in both cases (Table 1). BracketNet at the originally planned $\lambda=0.1$ behaves like the baselines.

200 **Closure regularization: large effects for SO(3), concentrated effects for T^2 .** For SO(3), closure
 201 falls from 0.6163 to 0.0250 (lower on 20/20 seeds), and the ground-truth distance falls from 0.4754
 202 to 0.2429 (19/20). Both are significant after Holm correction (Table 2).

203 For T^2 the closure test is also significant, but the effect is concentrated: the median paired difference
 204 is -8.9×10^{-5} , and the differences are heavily skewed (skewness -1.94). The t -interval includes
 205 zero, while the interval obtained by inverting the primary sign-flip test, $[-0.5331, -0.0067]$, does
 206 not. The difference comes from the few seeds in which +COMP. fails to close. The T^2 ground-truth
 207 distance does not change significantly.

208 Ten-step transport shows no detectable difference in either group; the 95% intervals are
 209 $[-0.0199, 0.0066]$ and $[-0.0080, 0.0061]$.

210 **Three SO(3) outcomes.** Every run falls into one of the classes of Prop. 2 or is non-closed (Fig. 1a).

- 211 • BracketNet: 8 correct (diagonal), 8 chiral, 4 non-closed.
- 212 • +COMP.: 5 correct, 0 chiral, 15 non-closed.

213 Closure converts non-closed runs into closed ones, and about half of them land in the wrong class.
 214 The rise in correct-closed runs is not significant (exact McNemar $p = 0.25$).

215 6.2 Mechanism: endpoint inference and chiral solutions

216 We tested Prop. 3 with predictions registered before any of the following runs.

217 **Composition error by class (existing runs).** As predicted, chiral runs compose far better than
 218 correct ones: median composition error 3.3×10^{-5} (chiral) vs. 7.9×10^{-4} (correct). Their one-step
 219 errors are the same (0.0035 vs. 0.0034). Their action disagreement with the truth is 1.01 vs. 0.38.
 220 The correct runs' value is the floor imposed by Prop. 3(c); chiral runs are no better than the identity.

221 **Closure geometry alone favours the correct class.** Pure closure flow from random subspaces
 222 reaches the diagonal class in 192/200 starts (Sec. 4), so the high chiral rate in training is not a property
 223 of the closure penalty alone.

224 **Trace of the generator span.** We re-ran the 20 BracketNet SO(3) runs with generator snapshots;
 225 20/20 reproduced the stored runs bit-for-bit. We measured each span's chirality $\chi = e_L - e_R$, the
 226 share of its energy in $\mathfrak{su}(2)_L$ minus that in $\mathfrak{su}(2)_R$.

- 227 • *Pre-registered.* At the start of the closure ramp, $|\chi|$ predicts a chiral outcome among the 16
 228 closed runs with AUC 0.75 (at initialization: 0.61). Medians are 0.22 (ending chiral) vs.
 229 0.13 (ending correct).
- 230 • *Post hoc.* The closure residual at the ramp start separates outcomes more sharply (AUC
 231 0.94). 7 of the 8 runs that end correct had already nearly closed onto the diagonal during the
 232 transition-fitting phase. 7 of the 8 chiral runs were still far from closed when the penalty
 233 switched on (Fig. 1b).

234 Together with the closure flow, this suggests the following picture. Transition fitting sometimes finds
 235 the diagonal algebra by itself, and closure then keeps it. When the span is still far from closed, closing
 236 it under the transport and composition losses, which admit exact chiral solutions at the ground-truth
 237 embedding but no exact continuous true-algebra solution (Prop. 3), usually yields a chiral algebra.

238 **Intervention: supplying the true action coefficients (pre-registered).** Prop. 3 predicts that chiral
 239 solutions lose their advantage if actions need not be inferred from endpoints. On 20 new seeds
 240 (400–419) we trained BracketNet with the true coefficients in the transport terms. This is the action-
 241 supervised setting of homomorphism autoencoders [Keurti et al., 2023]; q is fit only for evaluation.
 242 Results, relative to standard BracketNet on the same seeds:

- 243 • chiral outcomes fell from 6/20 to 0/20 (exact McNemar $p = 0.03$);
- 244 • correct-closed runs rose from 11/20 to 20/20 ($p = 3.9 \times 10^{-3}$);

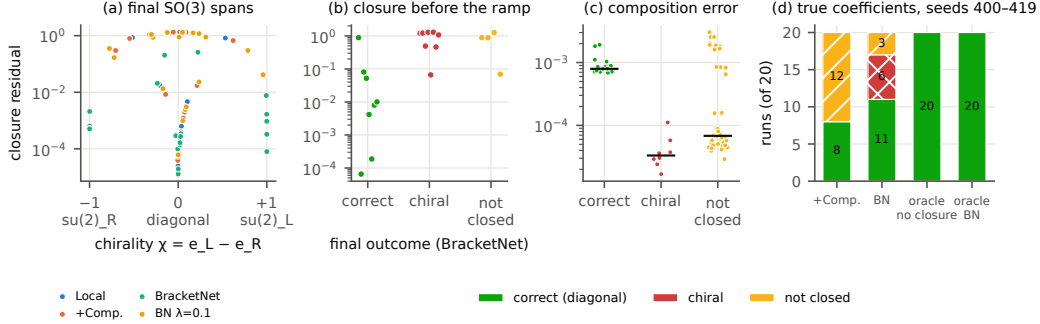


Figure 1: (a) Final SO(3) span chirality χ vs. closure residual for every fresh-seed run: closed runs sit at $\chi \in \{-1, 0, 1\}$, the classes of Prop. 2. (b) Closure of the BracketNet span when the closure penalty switches on, by final outcome (post hoc). (c) Composition error by outcome class, all SO(3) runs. (d) Outcomes with endpoint-inferred vs. true (oracle) action coefficients, seeds 400–419.

- the ground-truth distance changed by -0.215 ($[-0.332, -0.098]$);
- without any closure term, the oracle model also reaches 20/20 correct-closed runs.

Once actions are given, closure regularization is unnecessary: the difficulty lies in inferring actions from endpoints. The oracle models’ own q , which must infer actions from endpoints at test time, still has action disagreement 0.41, the floor of Prop. 3(c). The intervention adds information as well as removing the ambiguity, so it supports the mechanism but does not isolate it.

6.3 A rendered-image benchmark

To test a harder observation process, we rendered the SO(3) state as 20×20 images: the 3-D position of a Gaussian blob sets its image position and size, and the invariant coordinate sets its brightness ($p = 400$). The frozen protocol was applied unchanged on 20 new seeds (300–319), with predictions registered in advance.

- *No baseline run closes* (Local 20/20, +COMP. 20/20 non-closed). BracketNet lowers closure on 20/20 seeds (0.062 vs. 1.227, one-sided $p = 9.5 \times 10^{-7}$).
- *Every closed BracketNet run is chiral* (7/20; none diagonal). This is read off the chirality invariant, which needs no latent alignment.
- *Ground-truth recovery cannot be assessed by our alignment metric here*. The learned latents are far from an isometry of the true state (CKA with it 0.623, Procrustes residual 0.611).
- *There is a small transport cost*: 10-step error rises by 4.5×10^{-4} ($[1.4 \times 10^{-4}, 7.7 \times 10^{-4}]$).

Under rendering, closure regularization therefore produced closed algebras only of the wrong class. The benchmark is still synthetic and does not test natural images.

7 Representation alignment: agreement in embeddings vs. agreement in transformations

Models trained from different seeds can be compared at five levels: embedding geometry, transformation algebra, group representation, inferred action, and ground truth. We compare each pair of models on a shared probe set:

- *alignment*: the orthogonal Procrustes map R from $E_a(x)$ to $E_b(x)$, fitted on probe set A;
- *embedding agreement*: linear CKA;
- *algebra agreement*: the cross-seed algebra distance between $\text{span}(RA^{(a)}R^\top)$ and $\text{span}(A^{(b)})$;

274 • *transfer*: on a held-out probe set B , apply the action inferred by a in b 's space, $\hat{x}_{t+1} =$
 275 $D_b(R\rho_a R^\top E_b(x_t))$. The *transfer gap* is 0 when this is as good as b 's own action and 1
 276 when it is no better than the identity.

277 Independent units are 10 disjoint seed pairs, the same for every method (Appendix Table 5).

278 **Algebras agree more, through a few discrete classes.** For $SO(3)$, the cross-seed algebra distance
 279 falls from 0.6845 to 0.3922 (lower in 9/10 pairs, Holm $p = 8.8 \times 10^{-3}$). For T^2 it does not change
 280 significantly.

281 The $SO(3)$ pairs separate by class:

- 282 • both models correct: distance 0 (1 pair);
- 283 • both in the same *wrong* chiral class: distance 0 (1 pair);
- 284 • correct vs. chiral: distance $\approx \frac{1}{2}$, as fixed by Prop. 2 (4 pairs);
- 285 • at least one model non-closed: 4 pairs.

286 All 10 +COMP. pairs involve a non-closed model and spread over $[0, 1]$. Closure thus replaces
 287 continuous disagreement with disagreement between a few closed classes, so cross-model agreement
 288 does not certify correctness (Cor. 1).

289 **Transfer tracks structural agreement, not correctness.** Over all 190 pairs of each method (pooled,
 290 descriptive):

- 291 • $SO(3)$ pairs of two correct models transfer almost perfectly: median gap 0.013, 0% failures
 292 (gap > 0.5), over 63 pairs.
- 293 • Pairs of two models in the same wrong class also transfer: median gap 0.177, 0% failures.
 294 Aligning them through the ground truth instead of to each other breaks transfer (median gap
 295 4.151).
- 296 • Correct-vs-chiral pairs transfer worse: median gap 0.365, 19% failures.

297 What makes transformations reusable across models is that both models learned the *same* algebra,
 298 whether or not it is correct.

299 **CKA vs. structural distance as predictors of transfer failure (pre-registered; prediction not**
 300 **supported).** We pre-registered that the label-free algebra distance would predict transfer failure
 301 better than CKA. It did not.

- 302 • AUC for T^2 : CKA 1.00 vs. algebra distance 0.98.
- 303 • AUC for $SO(3)$: CKA 0.75 vs. algebra distance 0.62; difference interval $[-0.40, 0.30]$
 304 (seed-level cluster bootstrap).

305 Most transfer failures come from seed-level encoder failures, which CKA detects. CKA is, however,
 306 blind to *class* mismatches: $SO(3)$ correct-vs-chiral pairs have median CKA 0.989, the same as correct
 307 pairs (0.991), yet a far larger transfer gap. Embedding similarity and algebra agreement capture
 308 different failure modes, and a structural comparison needs both.

309 8 Ablations

310 Ablations use seeds 200–209 with same-seed references (Appendix Table 7).

- 311 • *Replication.* +COMP. again has higher $SO(3)$ closure and ground-truth distance than Bracket-
 312 Net (+0.2176, $[0.0786, 0.3567]$).
- 313 • *Anti-collapse term.* Essential: at weight 1 the T^2 latent distorts (CKA with the truth 0.73)
 314 and correct-closed runs fall to 0/10.
- 315 • *No detectable effect:* the fit–close–refit schedule (constant penalty or no freeze), the Gram
 316 penalty, and a span-only closure objective. We do not claim the schedule is necessary.
- 317 • *Misspecified K .* With $K+1$ no run is correct-closed (0/10 for $SO(3)$), as Prop. 2(c) requires.

9 Limitations

- **Synthetic scope.** Both benchmarks, including the rendered one, are synthetic, low-dimensional and compact, with known K and orthogonal actions. We make no claim about natural images, physical systems or large models.
- **No identifiability.** Wrong closed algebras are frequent for $\text{SO}(3)$, and correct algebras still have irreducible action ambiguity (Prop. 3c). Our ground-truth comparisons use orthogonal Procrustes, so they assume a near-isometric encoder; the T^2 failures are seed-level fitting failures shared by all methods.
- **Limited baselines.** We compare controlled objective variants and an action-supervised oracle; we do not reimplement published methods with different supervision.
- **Statistics and protocol.** We use 20 seeds; effects outside $\text{SO}(3)$ are small or concentrated in few seeds. No transport improvement is established; on rendered images transport is slightly worse. The closure-weight rule needed an amendment, committed before any fresh run. One pre-registered prediction (structure beats CKA at predicting transfer failure) was not supported.
- **Theory is local, qualitative and self-checked.** The escape bound covers only small actions; the classification is specific to $\mathfrak{so}(4)$ and $\mathfrak{so}(5)$; Prop. 3 concerns exact zeros at the ground-truth embedding, gives no quantitative gap and says nothing about the optimization path. The proofs have not been checked independently of the AI system that drafted them.

10 Conclusion

Closure turns continuous disagreement between transformation models into a choice among a few closed subalgebras. For $\text{SO}(3)$ the choice is often wrong, because endpoints cannot reveal rotations about the axis through a point. Agreement, of embeddings or of transformations, is not evidence of correctness.

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A Proofs

Invariance law. Under $B \mapsto BM$, $[A'_i, A'_j] = \sum_{k,l} M_{ki} M_{lj} [A_k, A_l]$, so the commutator matrix becomes $C(M \otimes M)$, while $P_0(BM) = P_0(B)$. Hence $R \mapsto R(M \otimes M)$, and the bounds follow from $\sigma(M \otimes M) = \{\sigma_i \sigma_j\}$. For $\varepsilon > 0$, $P_\varepsilon(BM) = BM(M^\top GM + \varepsilon I)^{-1} M^\top B^\top = B(G + \varepsilon M^{-\top} M^{-1})^{-1} B^\top$.

Proposition 1. (a) The commutator is bilinear and antisymmetric and satisfies Jacobi, so closure makes $\mathfrak{a}$ a Lie subalgebra; the Lie subgroup theorem [Hall, 2015, Ch. 5] gives the integrating subgroup. It need not be closed: an irrational line in a torus is an example.

(b) Write $X = \sum x_i A_i$, $Y = \sum y_j A_j$ and $r_{ij} = N[A_i, A_j]$. Then $\|N[X, Y]\| \leq \|x\| \|y\| (\sum \|r_{ij}\|^2)^{1/2} = \|x\| \|y\| K \sqrt{\mathcal{L}_{\text{close}}}$ by Cauchy-Schwarz, and $\|X\|^2 = x^\top G x \geq \lambda_{\min} \|x\|^2$.

(c) Dynkin's form [Bonfiglioli and Fulci, 2012] is $\log(e^X e^Y) = \sum_{m \geq 1} \frac{(-1)^{m-1}}{m} \sum \frac{[w]}{|w| \prod r_i! s_i!}$, summed over words $w = X^{r_1} Y^{s_1} \dots X^{r_m} Y^{s_m}$ with $r_i + s_i \geq 1$, where $[w]$ is right-nested.

- By submultiplicativity, $\|[w]\| \leq 2^{|w|-1} r^{|w|}$.
- Let e_n bound $\|N[w]\|$ for $|w| = n$, so $e_1 = 0$ and $e_2 \leq \delta r^2$. Write $[w] = [W, Z]$ with W a letter.
- $[W, P_0 Z]$ has both arguments in $\mathfrak{a}$, so its escape is at most $\delta r \|Z\| \leq \delta 2^{n-2} r^n$; and $\|[W, N Z]\| \leq 2r e_{n-1}$. By induction $e_n \leq (n-1) 2^{n-2} \delta r^n$.
- Each word therefore contributes at most $\frac{\delta}{4} (2r)^{|w|} / \prod r_i! s_i!$. The full sum is $\sum_m \frac{1}{m} (e^{4r} - 1)^m = -\ln(2 - e^{4r})$ for $r < \ln 2/4$.
- The length-1 words lie in $\mathfrak{a}$ and account for $4r$; subtracting them gives the bound.
- The length-2 words sum to $\frac{1}{2} [X, Y]$, whose escape is at most $\frac{1}{2} \delta r^2$; replacing their crude contribution $4\delta r^2$ gives $\frac{1}{2} \delta r^2 + O(\delta r^3)$.
- The series converges absolutely on this ball, and the principal logarithm is analytic there ($\|e^X e^Y - I\| < 1$). Both sides agree near 0, hence on the whole connected ball.

(d) With $d_t = \hat{z}_t - z_t$: $d_{t+1} = R_t d_t - e_t$ and $\|R_t d_t\| = \|d_t\|$. $\square$

Proposition 2. Subalgebras of $\mathfrak{so}(d)$ carry the ad-invariant negative-definite form $\text{tr}(XY)$, hence are compact and reductive.

(a) A 2-dimensional reductive algebra is abelian. It is maximal abelian because $\text{rank } \mathfrak{so}(5) = 2$, and maximal abelian subalgebras of a compact Lie algebra are conjugate [Adams, 1969].

(b) A 3-dimensional reductive $\mathfrak{h}$ is its center plus a semisimple part of dimension 0 or 3. It cannot be abelian (the rank is 2), so $\mathfrak{h} \cong \mathfrak{su}(2)$. Its projections onto the ideals $\mathfrak{su}(2)_L, \mathfrak{su}(2)_R$ are zero or injective, because $\mathfrak{h}$ is simple.

- If one projection is zero, $\mathfrak{h}$ is a factor. The factors are ideals, hence invariant under all of $SO(4)$, and a reflection exchanges them.
- Otherwise $\mathfrak{h}$ is the graph of an isomorphism $\mathfrak{su}(2)_L \rightarrow \mathfrak{su}(2)_R$. Any two such graphs differ by an automorphism of $\mathfrak{su}(2)$, which is inner, and $SU(2) \times SU(2) \rightarrow SO(4)$ is onto, so all such $\mathfrak{h}$ are $SO(4)$ -conjugate. The stabilizer of $e_4, \mathfrak{so}(3)$ on $\mathbb{R}^3 \oplus \mathbb{R}$, is one of them (it is not an ideal).
- $SO(4)$ is connected, so it cannot exchange the two ideals; hence three classes under $SO(4)$ and, since a reflection fixing e_4 exchanges them and preserves the diagonal, two under $O(4)$.

The Pfaffian is $SO(4)$ -invariant and odd under reflections. It vanishes on rank-2 rotations and equals $\pm \frac{1}{2}$ on unit elements of $\mathfrak{su}(2)_{L/R}$.

(c) A 4-dimensional reductive subalgebra is $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$ with the $\mathfrak{u}(1)$ central. The centralizer of the diagonal $\mathfrak{so}(3)$ is trivial, because $\mathfrak{so}(4) \cong \Lambda^2 \mathbb{R}^4$ decomposes as $\mathbf{3} \oplus \mathbf{3}$ under it, with no trivial summand; the same holds for its conjugates. So the $\mathfrak{su}(2)$ is a factor, which equals the derived

458 algebra $[\mathfrak{h}, \mathfrak{h}]$. A diagonal $\mathfrak{so}(3) \subset \mathfrak{h}$ would satisfy $\mathfrak{so}(3) = [\mathfrak{so}(3), \mathfrak{so}(3)] \subset [\mathfrak{h}, \mathfrak{h}]$, a factor, which is
 459 impossible.

460 *Scope.* The classification is for exactly closed spans. Learned spans are only approximately closed;
 461 we assign them to classes using continuous invariants (Pfaffian, chirality) and the closure-flow check,
 462 without a stability theorem relating approximate closure to distance from a closed class. $\square$

463 **Proposition 3.** Here $g \in SO(3)$ acts on $z_{1:3}$ and fixes z_4 , so transitions preserve $\|z\|$.

464 (a) Identify $\mathbb{R}^4$ with the quaternions $\mathbb{H}$. The exponential of $\mathfrak{su}(2)_L$ is left multiplication by unit
 465 quaternions, which acts simply transitively on each sphere $\|z\| = \rho > 0$. For $\|z\| = \|z'\|$, $u = z'z^{-1}$
 466 is the unique unit quaternion with $L_u z = z'$.

- 467 • $q = \log u$ is smooth wherever $u \neq -1$, and $\exp(\log u) = u$ for every branch.
- 468 • $u_{02} = z_2 z_0^{-1} = (z_2 z_1^{-1})(z_1 z_0^{-1}) = u_{12} u_{01}$, so composition is exact.
- 469 • Closure is exact, the basis is orthonormal so the Gram penalty is 0, and $\text{Cov}(z) = I$ in
 470 population (states are $\mathcal{N}(0, I)$ and rotations preserve that law).

471 The coefficient of a rotation by angle $\theta = \angle(z, z')$ has entries up to $\sqrt{2}\theta$; only pairs beyond the
 472 bound 0.65 break exactness.

473 (b) Fix the sphere S and suppose a continuous rule gives $R(z, z') = \rho(q(z, z')) \in SO(3)$ with
 474 $R(z, z')z = z'$ and $R(z_0, z_2) = R(z_1, z_2)R(z_0, z_1)$ for all $z_i \in S$ with pairwise angles $< \epsilon$.

- 475 • The triple (z, z, z) gives $R(z, z) = R(z, z)^2$, so $R(z, z) = I$.
- 476 • For a chain $x_0, \dots, x_n$ with consecutive angles $< \epsilon/3$, let $P = R(x_{n-1}, x_n) \cdots R(x_0, x_1)$.
 477 Inserting or deleting a point inside an ϵ -small triangle leaves P unchanged. Two chains with
 478 the same endpoints and pointwise angles $< \epsilon/3$ are joined by such moves (replace one point
 479 at a time; each move is a triangle of diameter $< \epsilon$), so P is unchanged. Fine subdivisions of
 480 a continuous path therefore define one P , and by uniform continuity it is constant along a
 481 homotopy of paths with fixed endpoints.
- 482 • $S \cong S^2$ is simply connected, so fixing $e \in S$, $s(z) = P(e \rightarrow z)$ is well defined, with
 483 $s(z)e = z$. It is continuous, because $s(z') = R(z, z')s(z)$ for nearby z' and R is continuous
 484 with $R(z, z) = I$.
- 485 • For a unit $f \perp e$ (in the $\mathbb{R}^3$ part), $z \mapsto s(z)f$ is orthogonal to $s(z)e = z$: a nowhere-
 486 vanishing tangent field on S^2 , contradicting the hairy-ball theorem.

487 The rule takes values in the true (diagonal) group; the argument uses no property of q beyond
 488 continuity.

489 (c) For fixed endpoints (z, z') with $z_{1:3} \neq 0$, the set $\{g : gz = z'\}$ is a coset of the circle stabilizing
 490 z . The states have a density and $\exp$ pushes the Gaussian coefficients to a law with a density on
 491 $SO(3)$, so the conditional law of g on that circle has a density for almost every (z, z') . Hence
 492 $\mathbb{E}\|\hat{g} - g\|^2 \geq \mathbb{E} \text{tr Cov}(g \mid z, z') > 0$ for every endpoint estimator $\hat{g}$. $\square$

493 **Remark (what Prop. 3b does not show).**

- 494 • *Continuity is essential.* S minus one point is contractible, so $SO(3) \rightarrow S^2$ has a continuous
 495 section s there. Ignoring the coefficient bound, $R(z, z') = s(z')s(z)^{-1}$ is a measurable rule
 496 that is exact on every triple avoiding that point, i.e. off a null set.
- 497 • *A gap for any fixed Lipschitz class.* Let $J_\epsilon(q)$ be the expected transport and composition
 498 loss on data triples with pairwise angles $< \epsilon$, small enough that the chiral rule is within the
 499 bound ($J_\epsilon = 0$ for it). For every L , the infimum of J_ϵ over L -Lipschitz rules with bounded
 500 coefficients is positive for the true algebra. Otherwise, by Arzelà–Ascoli a subsequence
 501 converges locally uniformly to such a rule with $J_\epsilon = 0$, by dominated convergence (bounded
 502 integrand). The integrand is continuous and the support contains every such triple on every
 503 orbit sphere, so the rule is exact there, which contradicts (b).
- 504 • *No quantitative statement.* We do not bound this gap, and it may vanish as $L \rightarrow \infty$. Nor do
 505 we compare the full training objective, which includes large steps and clipping, between the
 506 two algebras.

507 **Corollary 1.** This is Prop. 3(a) applied to both models, together with Prop. 2(b) and the distance $\frac{1}{2}$
508 between the diagonal and a chiral span. $\square$

509 B Protocol details

510 **Architecture and optimization.** E, D, q are two-hidden-layer SiLU MLPs of width 64. The latent
511 dimension is d , generators are initialized as $W \sim \mathcal{N}(0, 1/d)$, and coefficients are $0.65 \tanh(\cdot)$.
512 Training uses AdamW (lr 2×10^{-3} , weight decay 10^{-5}), batch 160, gradient clipping at 5, $\varepsilon = 10^{-6}$,
513 Gram weight 1 and anti-collapse weight 16. The anti-collapse weight was chosen in a preliminary
514 round on seeds 5–6 by a method-agnostic criterion. The final iterate is evaluated, with no early
515 stopping. All runs are CPU-only and bit-identical on repeat.

516 Order of operations.

- 517 1. Selection rules committed.
- 518 2. Development grid (280 runs).
- 519 3. Rules applied by script.
- 520 4. Fresh seeds 100–119 and ablation seeds 200–209, each run once.
- 521 5. Review predictions committed.
- 522 6. Trace, oracle and rendered runs.

523 No setting was changed after any fresh-seed result was seen.

524 **Rendered benchmark.** The image has 20×20 pixels on $[-3.2, 3.2]^2$. The blob centre is (z_1, z_2) ,
525 its width is $0.45 e^{0.3z_3}$ and its amplitude is $1 + 0.5 \tanh z_4$. Positions are scaled by 0.7 so that blobs
526 stay in view. Centre, width and amplitude determine z , so the map is injective.

527 **Oracle intervention.** The transport terms use the true coefficients of each training edge. The
528 composition term is dropped, because a_{02} is a product rather than a single coefficient. q is trained
529 only by regression to the true coefficients, for evaluation.

530 C Statistical audit

531 The six primary tests were recomputed from the raw run files by an independent implementation, and
532 all matched. The t -interval can disagree with the primary sign-flip test when differences are skewed,
533 so we also report the interval obtained by inverting the two-sided exact sign-flip test, which matches
534 the primary procedure, and a percentile bootstrap interval.

Table 3: Paired differences (BracketNet — +Comp.). Intervals: t ; sign-flip-inverted; bootstrap.

Test	Group	n	median diff.	skew	t 95%	inverted sign-flip 95%	bootstrap 95%
H1 closure	T^2	20	-0.0001	-1.94	[-0.5427, +0.0049]	[-0.5331, -0.0067]	[-0.5410, -0.0380]
H2 GT distance	T^2	20	-0.0000	-2.58	[-0.0674, +0.0240]	[-0.0659, +0.0172]	[-0.0688, +0.0136]
H3 cross-seed	T^2	10	-0.0001	-0.33	[-0.1421, +0.1106]	[-0.1410, +0.1091]	[-0.1217, +0.0862]
H1 closure	SO(3)	20	-0.6501	-0.18	[-0.8550, -0.3278]	[-0.8566, -0.3269]	[-0.8295, -0.3572]
H2 GT distance	SO(3)	20	-0.3321	0.21	[-0.3348, -0.1301]	[-0.3347, -0.1293]	[-0.3231, -0.1378]
H3 cross-seed	SO(3)	10	-0.3466	0.30	[-0.4424, -0.1423]	[-0.4464, -0.1439]	[-0.4137, -0.1664]

535 **Threshold sensitivity (robustness check, not pre-registered).** SO(3) fresh-seed counts (correct
536 / chiral / other closed / non-closed) under alternative thresholds, as a fraction of the random-span
537 closure level and of the chance distance:

538

closed	correct	+Comp.	BracketNet
0.1%	5%	3/0/0/17	8/5/0/7
0.1%	10%	3/0/0/17	8/5/0/7
0.1%	20%	3/0/0/17	8/5/0/7
1%	5%	5/0/0/15	8/8/0/4
1%	10%	5/0/0/15	8/8/0/4
1%	20%	5/0/0/15	8/8/0/4
5%	5%	7/0/0/13	8/9/1/2
5%	10%	7/0/0/13	9/9/0/2
5%	20%	7/0/0/13	9/9/0/2

539 The qualitative conclusions (no chiral solutions without closure; several with it; more correct-closed
540 runs with closure) hold at every setting.

541 D Comparison of settings

Work	Supervision	Learned structure	Closure	Cross-model structure
Rao & Ruderman; Dickstein et al.	transformed pairs	linear generators	not enforced	no
Cohen & Welling 2014	pairs, commutative group	irreps of a torus	abelian by design	no
LieGAN / LaLiGAN	distributional invariance	generators (latent)	not penalized	no
Forestano et al. 2023	labelled data	generators	closure loss	no
Quessard et al.; Painter et al.	interaction (Painter: unlabelled)	rotation representations	$SO(n)$ by design	no
Homomorphism AE	transitions + action labels	homomorphism into matrix group	via composition	no
CKA / shape metrics	trained networks	—	—	embeddings only
This work	transitions, no labels	generators + endpoint actions	closure penalty	algebras, transfer

542 These summaries describe each setting at the level needed to judge overlap, from abstracts and prior
543 knowledge (the full texts could not be retrieved); see the cited papers for details.

544 E Additional results

Table 4: Fresh-seed results for all groups, including SO(2) (closure identically 0 for one generator).

Group	Method	1-step MSE	10-step MSE	Closure residual	GT dist.	CC
SO(2)	Local	0.0026 ± 0.0010	0.0061 ± 0.0019	0 ± 0	0.000 ± 0.000	20/20
SO(2)	+Comp.	0.0026 ± 0.0010	0.0065 ± 0.0028	0 ± 0	0.000 ± 0.000	20/20
SO(2)	BracketNet	0.0026 ± 0.0010	0.0063 ± 0.0027	0 ± 0	0.000 ± 0.000	20/20
SO(2)	BracketNet ($\lambda=0.1$)	0.0026 ± 0.0010	0.0063 ± 0.0027	0 ± 0	0.000 ± 0.000	20/20
T^2	Local	0.0080 ± 0.0076	0.0645 ± 0.0841	$3.4 \times 10^{-1} \pm 7.2 \times 10^{-1}$	0.161 ± 0.282	14/20
T^2	+Comp.	0.0081 ± 0.0076	0.0605 ± 0.0757	$2.9 \times 10^{-1} \pm 6.3 \times 10^{-1}$	0.151 ± 0.274	14/20
T^2	BracketNet	0.0072 ± 0.0079	0.0539 ± 0.0792	$2.2 \times 10^{-2} \pm 5.6 \times 10^{-2}$	0.129 ± 0.282	16/20
T^2	BracketNet ($\lambda=0.1$)	0.0082 ± 0.0078	0.0622 ± 0.0785	$3.1 \times 10^{-1} \pm 6.1 \times 10^{-1}$	0.157 ± 0.277	14/20
SO(3)	Local	0.0048 ± 0.0027	0.0325 ± 0.0312	$6.3 \times 10^{-1} \pm 5.7 \times 10^{-1}$	0.479 ± 0.435	6/20
SO(3)	+Comp.	0.0048 ± 0.0031	0.0328 ± 0.0299	$6.2 \times 10^{-1} \pm 5.7 \times 10^{-1}$	0.475 ± 0.433	5/20
SO(3)	BracketNet	0.0049 ± 0.0034	0.0319 ± 0.0341	$2.5 \times 10^{-2} \pm 7.1 \times 10^{-2}$	0.243 ± 0.254	8/20
SO(3)	BracketNet ($\lambda=0.1$)	0.0050 ± 0.0031	0.0484 ± 0.0484	$5.4 \times 10^{-1} \pm 5.4 \times 10^{-1}$	0.453 ± 0.413	5/20

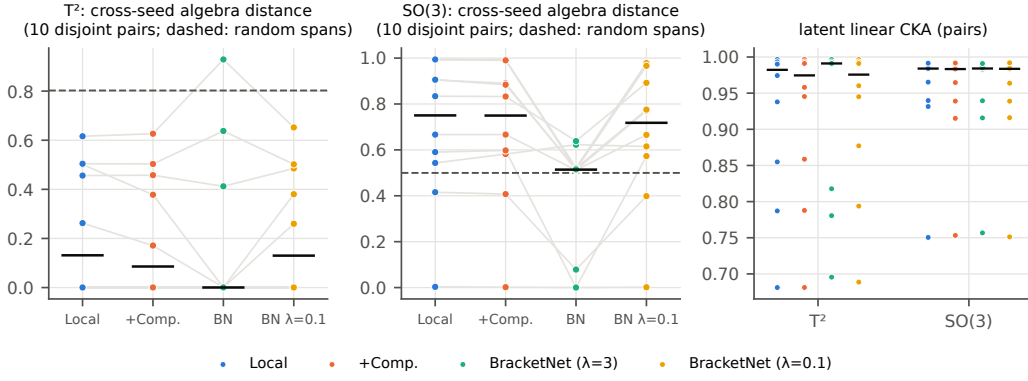


Figure 2: Cross-seed algebra distance on 10 disjoint pairs (grey lines join the same pair across methods; BN = BracketNet) and latent CKA.

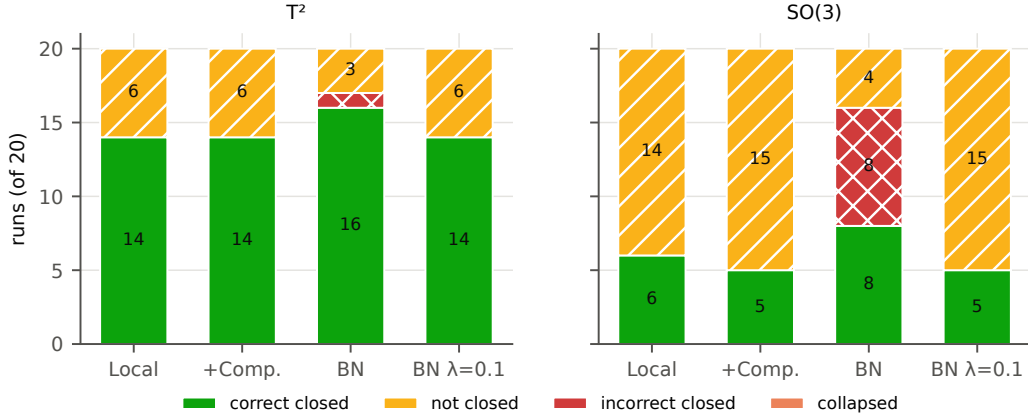


Figure 3: Structural category of each fresh-seed run (BN = BracketNet).

545 F Reproducibility

546 The anonymized supplement contains the data generator, models, training and analysis code, the
 547 selection rules and review predictions, per-run result files for every run, the fresh-seed checkpoints,

Table 5: Cross-seed alignment on 10 disjoint pairs (mean $\pm$ s.d.). Algebra distance chance: 0.80 (T^2), 0.50 (SO(3)). Transfer gap: 0 = native, 1 = identity. Projected: transferred actions re-expressed in b 's generators.

Group	Method	CKA	Algebra dist.	Action disagr.	Transfer gap	Projected gap
T^2	Local	0.920 $\pm$ 0.110	0.234 $\pm$ 0.261	0.596 $\pm$ 0.662	0.876 $\pm$ 1.621	0.647 $\pm$ 1.069
T^2	+Comp.	0.920 $\pm$ 0.109	0.214 $\pm$ 0.252	0.580 $\pm$ 0.659	0.852 $\pm$ 1.573	0.626 $\pm$ 1.032
T^2	BracketNet	0.925 $\pm$ 0.114	0.198 $\pm$ 0.341	0.485 $\pm$ 0.785	0.769 $\pm$ 1.406	0.432 $\pm$ 0.838
T^2	BracketNet ($\lambda=0.1$)	0.923 $\pm$ 0.106	0.228 $\pm$ 0.259	0.592 $\pm$ 0.662	0.770 $\pm$ 1.398	0.611 $\pm$ 0.997
SO(3)	Local	0.951 $\pm$ 0.074	0.685 $\pm$ 0.312	2.280 $\pm$ 0.995	0.745 $\pm$ 1.033	0.923 $\pm$ 0.413
SO(3)	+Comp.	0.949 $\pm$ 0.073	0.685 $\pm$ 0.308	2.222 $\pm$ 0.973	0.741 $\pm$ 1.008	0.909 $\pm$ 0.393
SO(3)	BracketNet	0.952 $\pm$ 0.073	0.392 $\pm$ 0.257	2.015 $\pm$ 0.815	0.732 $\pm$ 0.919	1.257 $\pm$ 0.536
SO(3)	BracketNet ($\lambda=0.1$)	0.949 $\pm$ 0.074	0.664 $\pm$ 0.296	2.194 $\pm$ 0.949	0.726 $\pm$ 0.978	0.924 $\pm$ 0.375

Table 6: Structural breakdown of fresh-seed runs (counts out of 20). Incorrect-closed runs are shown with the chiral count in parentheses. “Bad action” counts correct-closed runs whose action disagreement with the truth exceeds 0.5. PR: participation ratio.

Group	Method	Correct	Incorrect (chiral)	Non-closed	Collapsed	Bad action	Action disagr.	PR
T^2	Local	14	0 (0)	6	0	0	0.336 $\pm$ 0.438	2.98 $\pm$ 0.38
T^2	+Comp.	14	0 (0)	6	0	0	0.326 $\pm$ 0.429	2.93 $\pm$ 0.30
T^2	BracketNet	16	1 (0)	3	0	0	0.261 $\pm$ 0.405	2.86 $\pm$ 0.08
T^2	BracketNet ($\lambda=0.1$)	14	0 (0)	6	0	0	0.331 $\pm$ 0.433	2.93 $\pm$ 0.26
SO(3)	Local	6	0 (0)	14	0	0	0.849 $\pm$ 0.355	2.26 $\pm$ 0.44
SO(3)	+Comp.	5	0 (0)	15	0	0	0.838 $\pm$ 0.361	2.27 $\pm$ 0.44
SO(3)	BracketNet	8	8 (8)	4	0	0	0.764 $\pm$ 0.320	2.91 $\pm$ 1.00
SO(3)	BracketNet ($\lambda=0.1$)	5	0 (0)	15	0	0	0.833 $\pm$ 0.356	2.33 $\pm$ 0.46

figure scripts and unit tests. Every number in the text is generated as a $\LaTeX$ macro from the raw files.

G Use of AI assistance

Code, experiments, analyses and drafts of this manuscript were developed with substantial assistance from an AI coding assistant. The proofs were checked by the same AI system and by automated numerical tests; this self-review is not independent human verification. The authors are responsible for all content.

Table 7: Ablations on seeds 200–209 (mean $\pm$ s.d.). Coverage is the fraction of the true span contained in the learned span. Categories: correct / incorrect / non-closed / collapsed. With misspecified K , “correct” is impossible by definition.

Group	Variant	10-step MSE	Closure	GT dist.	Coverage	Categories
T^2	BracketNet (reference)	0.0157 $\pm$ 0.0047	$3.6 \times 10^{-5} \pm 2.0 \times 10^{-5}$	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	10/0/0/0
T^2	+Comp. (reference)	0.0231 $\pm$ 0.0275	$2.0 \times 10^{-1} \pm 6.2 \times 10^{-1}$	0.050 $\pm$ 0.156	0.950 $\pm$ 0.156	9/0/1/0
T^2	constant penalty, no staging	0.0370 $\pm$ 0.0475	$1.2 \times 10^{-4} \pm 2.2 \times 10^{-4}$	0.076 $\pm$ 0.170	0.924 $\pm$ 0.170	8/2/0/0
T^2	staged ramp, no freeze	0.0156 $\pm$ 0.0048	$3.4 \times 10^{-5} \pm 2.1 \times 10^{-5}$	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	10/0/0/0
T^2	no Gram penalty	0.0274 $\pm$ 0.0382	$1.4 \times 10^{-2} \pm 4.5 \times 10^{-2}$	0.027 $\pm$ 0.084	0.973 $\pm$ 0.084	9/0/1/0
T^2	anti-collapse weight 1	0.1997 $\pm$ 0.0253	$3.1 \times 10^{-3} \pm 4.6 \times 10^{-3}$	0.876 $\pm$ 0.130	0.124 $\pm$ 0.130	0/8/2/0
T^2	span-only (whitened) closure	0.0157 $\pm$ 0.0047	$3.6 \times 10^{-5} \pm 2.0 \times 10^{-5}$	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	10/0/0/0
T^2	0.3 λ	0.0236 $\pm$ 0.0287	$1.4 \times 10^{-1} \pm 4.5 \times 10^{-1}$	0.040 $\pm$ 0.127	0.960 $\pm$ 0.127	9/0/1/0
T^2	3 λ	0.0156 $\pm$ 0.0044	$1.6 \times 10^{-5} \pm 1.1 \times 10^{-5}$	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	10/0/0/0
T^2	$K+1$ generators	0.0248 $\pm$ 0.0194	$3.2 \times 10^{-1} \pm 1.1 \times 10^{-1}$	0.375 $\pm$ 0.103	0.937 $\pm$ 0.154	0/1/9/0
T^2	+Comp., $K+1$	0.0222 $\pm$ 0.0245	$8.2 \times 10^{-1} \pm 1.9 \times 10^{-1}$	0.367 $\pm$ 0.108	0.949 $\pm$ 0.161	0/0/10/0
T^2	$K-1$ generators	0.1917 $\pm$ 0.0300	0 $\pm$ 0	0.501 $\pm$ 0.001	0.499 $\pm$ 0.001	0/10/0/0
SO(3)	BracketNet (reference)	0.0218 $\pm$ 0.0160	$4.2 \times 10^{-2} \pm 8.0 \times 10^{-2}$	0.244 $\pm$ 0.298	0.756 $\pm$ 0.298	5/2/3/0
SO(3)	+Comp. (reference)	0.0393 $\pm$ 0.0309	$7.0 \times 10^{-1} \pm 5.2 \times 10^{-1}$	0.462 $\pm$ 0.412	0.538 $\pm$ 0.412	3/0/7/0
SO(3)	constant penalty, no staging	0.0298 $\pm$ 0.0394	$1.4 \times 10^{-2} \pm 2.6 \times 10^{-2}$	0.185 $\pm$ 0.298	0.815 $\pm$ 0.298	6/1/3/0
SO(3)	staged ramp, no freeze	0.0299 $\pm$ 0.0191	$2.6 \times 10^{-2} \pm 4.6 \times 10^{-2}$	0.239 $\pm$ 0.301	0.761 $\pm$ 0.301	5/2/3/0
SO(3)	no Gram penalty	0.0181 $\pm$ 0.0136	$1.5 \times 10^{-2} \pm 4.3 \times 10^{-2}$	0.177 $\pm$ 0.260	0.823 $\pm$ 0.260	6/2/2/0
SO(3)	anti-collapse weight 1	0.0302 $\pm$ 0.0140	$5.6 \times 10^{-2} \pm 2.2 \times 10^{-2}$	0.619 $\pm$ 0.063	0.381 $\pm$ 0.063	0/1/9/0
SO(3)	span-only (whitened) closure	0.0182 $\pm$ 0.0138	$4.0 \times 10^{-2} \pm 7.7 \times 10^{-2}$	0.243 $\pm$ 0.298	0.757 $\pm$ 0.298	5/2/3/0
SO(3)	0.3 λ	0.0314 $\pm$ 0.0284	$2.2 \times 10^{-1} \pm 3.4 \times 10^{-1}$	0.299 $\pm$ 0.297	0.701 $\pm$ 0.297	4/1/5/0
SO(3)	3 λ	0.0274 $\pm$ 0.0244	$1.7 \times 10^{-2} \pm 3.0 \times 10^{-2}$	0.237 $\pm$ 0.302	0.763 $\pm$ 0.302	5/2/3/0
SO(3)	$K+1$ generators	0.0225 $\pm$ 0.0108	$1.2 \times 10^{-1} \pm 2.4 \times 10^{-1}$	0.457 $\pm$ 0.076	0.723 $\pm$ 0.102	0/7/3/0
SO(3)	+Comp., $K+1$	0.0156 $\pm$ 0.0087	$5.5 \times 10^{-1} \pm 2.6 \times 10^{-1}$	0.511 $\pm$ 0.220	0.652 $\pm$ 0.293	0/0/10/0
SO(3)	$K-1$ generators	0.1253 $\pm$ 0.0535	$6.0 \times 10^{-1} \pm 5.1 \times 10^{-1}$	0.464 $\pm$ 0.166	0.536 $\pm$ 0.166	0/4/6/0

Table 8: Development grid (seeds 5–9; selection only; means over 5 seeds). For each method: 10-step MSE, closure, correct-closed count.

Group	Updates	MSE	Local closure	CC	MSE	+Comp. closure	CC	MSE	BracketNet closure	CC
T^2	420	0.1635	0.1374	1/5	0.1604	0.1104	1/5	0.1647	0.0600	1/5
T^2	1000	0.0704	0.0223	4/5	0.0678	0.0188	4/5	0.0676	0.0011	4/5
T^2	2000	0.0376	0.0016	4/5	0.0391	0.0022	4/5	0.0390	0.0005	4/5
T^2	4000	0.0308	0.0004	4/5	0.0312	0.0003	4/5	0.0317	0.0001	4/5
SO(3)	420	0.1391	0.8088	0/5	0.1532	0.7579	0/5	0.1479	0.3826	0/5
SO(3)	1000	0.0713	0.8762	1/5	0.0843	0.7771	1/5	0.0668	0.2032	1/5
SO(3)	2000	0.0396	0.9245	1/5	0.0308	0.6989	1/5	0.0573	0.1129	2/5
SO(3)	4000	0.0295	0.9574	1/5	0.0276	0.7086	1/5	0.0340	0.0568	2/5

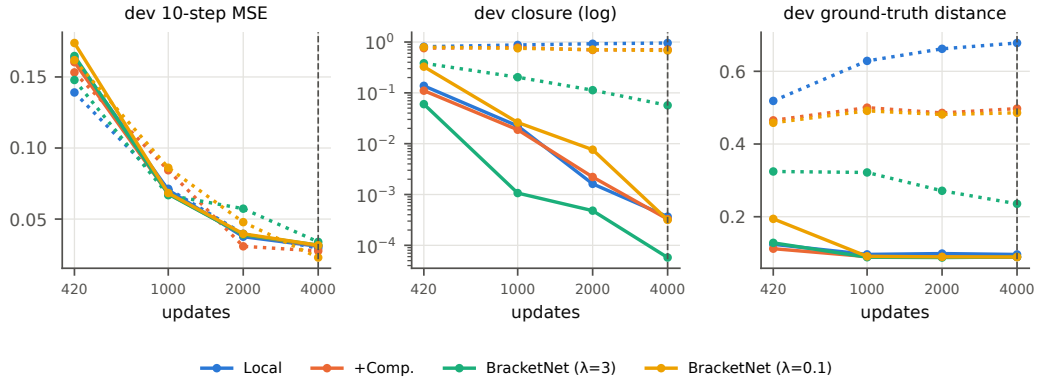


Figure 4: Development-seed budget sweep (solid: T^2 ; dotted: SO(3)); the vertical line marks the selected budget.

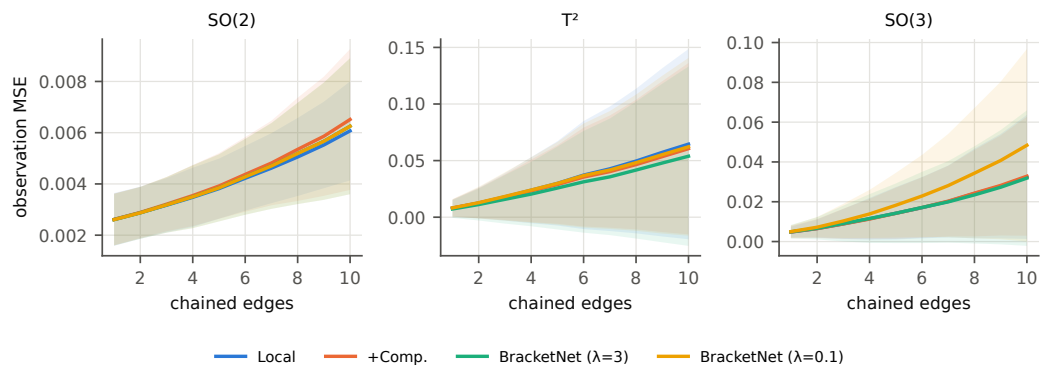


Figure 5: Chained transport error (mean $\pm$ s.d., 20 fresh seeds).